Accurate Response Models Do Not Necessarily Improve Recovery: A Native Temporal-Contract Component Study

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Abstract

We separate response-model accuracy from recovery-action selection in a native two-process timing component. Public pre-reference velocity observations identify both inertia and damping in the restricted noiseless rotor family. This removes eight erroneous conditional admissions but leaves all paired physical verdicts unchanged: fixed and identified full-branch control each yield 33/180 joint TRUE outcomes. Success-only retry yields 36/180, repairing four cases and harming one. A separately frozen developmental extension prefers a fully supported recovery before relaxing to a nominal retry; it yields 37/180, with four repairs and no harms relative to full-branch control. The study includes 612 new rollouts and 108 retained comparators. These results support stronger deterministic baselines, not learned-robot or chemical gains.

1. Separate model error and action priority

The preceding shared-stop study showed that stopping A on time can interrupt a required B process. A requires completion $T_A - \tau_A \in [0, U_A]$; B requires $T_B - \tau_B \in [L_B, U_B]$ and uninterrupted operation before its minimum. A candidate checks both processes along a finite retry-success, retry-failure and relay-fallback graph. This support check is conditional on the response model; it does not certify observed success.

We ask whether fixed-model error explains remaining failures. In the contact-free implicit-Euler rotor family, adjacent powered speeds satisfy

$$\begin{aligned} v_{k+1} &= \alpha v_k + \beta, \\ I &= u\Delta t\,\alpha/\beta, \quad d = I(1/\alpha - 1)/\Delta t. \end{aligned} \tag{1}$$

Ordinary least squares uses only adjacent delivered velocity/power samples. The fit freezes at A's first delivered reference, before any A stop request. It requires eight pairs, excitation, small residual and a declared parameter domain. Unsupported fits retain the fixed model. Four corners of a relative 10^{-5} numerical neighborhood supply response predictions; this is not statistically calibrated coverage. B's model remains unchanged.

2. Matched developmental execution

All controls share public timestamps, command reservations, A/B windows and the same non-preemptive selective channel. A relay stops both rotors. ACK indicates receipt; delivered power feedback reveals failure. True parameters and fault labels are withheld from the actor.

The first release executes 432 new trajectories: 216 on historical cases and 216 across two new inertia/damping settings, with 108 original fixed comparators retained. It follows eight focused tests and an explicit source, protocol and dependency freeze. New settings were selected during development; they are not blinded holdouts or novel task graphs.

The initial three arms use fixed full-branch control, public-ID full-branch control and public-ID success-branch retry. The latter admits nominally eligible retries even when their failure continuation lacks joint support. After observing an avoidable harmed case, we separately freeze an additional 180-run control: prefer any fully supported candidate; otherwise retain the success-only retry opportunity. No earlier run or control file is rewritten.

Across both releases there are 612,612 native samples and 6,732 recorded RGB frames; actors do not read RGB. A separate event oracle and native actuation reconstruction audit every new trajectory, with zero native discrepancy. Public replay reproduces all 612 controls. Separate least-squares reconstruction checks 540 fitted responses against withheld truth post hoc, with maximum absolute parameter error 9.1×10^{-17}. This precision is specific to the noise-free known family, not hardware validation. All 180 four-arm groups share initial mechanical prefixes.

Development increment in the ICML 2026 template; not a complete submission. [1]Research draft; author information pending. Correspondence to: Author contact details pending.

Preprint. October 12, 2026.

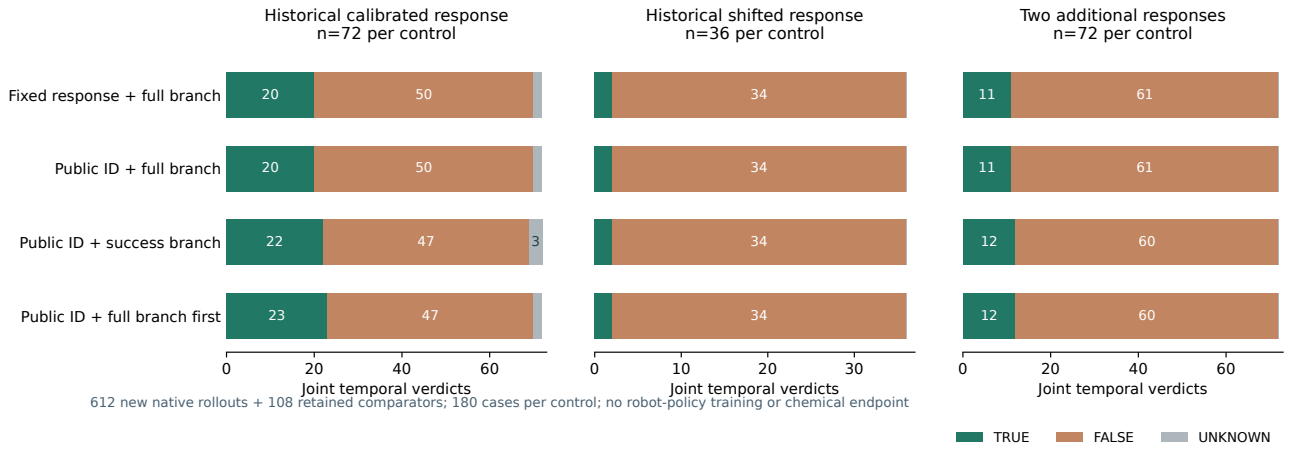


Figure 1. All 180 matched cases per control, including FALSE and UNKNOWN. The fixed control reuses 108 earlier trajectories and adds 72 new response cases. Both ID controls add 180 each; the full-branch-first extension adds another 180 after the initial three-arm audit. Repeated deterministic configurations are not independent tasks or chemistry batches.

Table 1. Joint timing verdicts over 180 cases per control.

Control	TRUE	FALSE	UNKNOWN
Fixed full branch	33	145	2
ID full branch	33	145	2
ID success branch	36	141	3
ID full branch first	37	141	2

3. Null gains and avoidable harms

Public ID reduces selected full-support/non-TRUE cases from eight to zero, yet repairs no physical outcome and harms none relative to fixed full-branch control. Correct prediction alone does not expand the action graph or make an infeasible deadline feasible.

Success-only retry repairs four first-drop cases but harms one persistent-fault case, changing its TRUE verdict to UNKNOWN. That case already has a fully supported deferred relay. It therefore exposes action-priority error, not an unavoidable benefit of learning. Full-branch-first admission repairs the same four cases and avoids that harm. All B continuity verdicts are TRUE; the extension still has 141 FALSE and two UNKNOWN joint outcomes.

For 240 matched first-drop/persistent-fault pairs, native and public histories are identical before the retry effect becomes observable. No controller uses the eventual fault label to choose that first retry. This constructed observational ambiguity is not a general safety theorem.

4. Scope and next discriminating experiment

Policy-weight training, physical robot trials and chemical endpoint evidence remain zero. The extension is selected after the initial results and is developmental, not confirmatory. Host work does not advance simulation; declared transport/actuation delays do. Numerical support is not guaranteed under sensor noise, model-family mismatch or hardware latency.

Before a VLA/agentic learning claim, the next baseline must quantify attainable performance under the current action graph and test anticipatory public scheduling. Learned context or weight absorption should then be compared with those controls using matched information and budgets. Recovery priority, system identification and learned skill acquisition require separate evidence. A post-run auditor correction only resolves old relative paths against their producer; its diff and red/green regression evidence are retained.